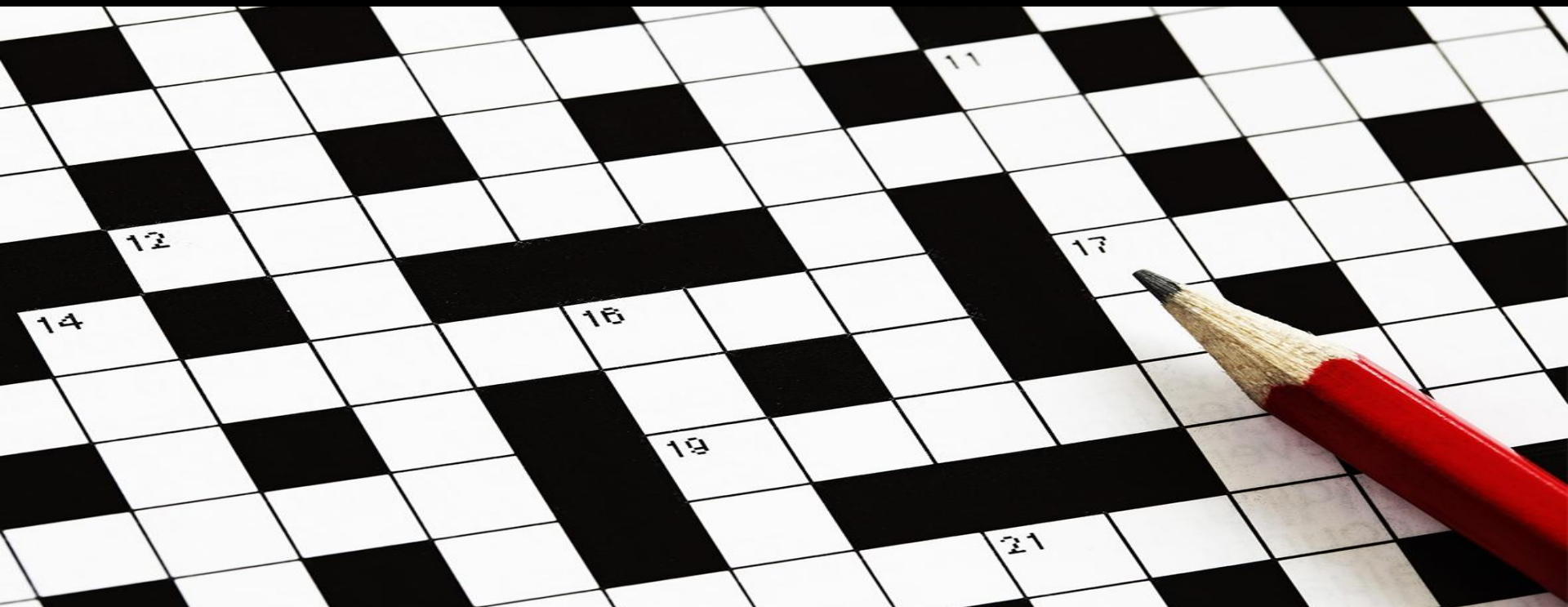


Human Defence Responses *

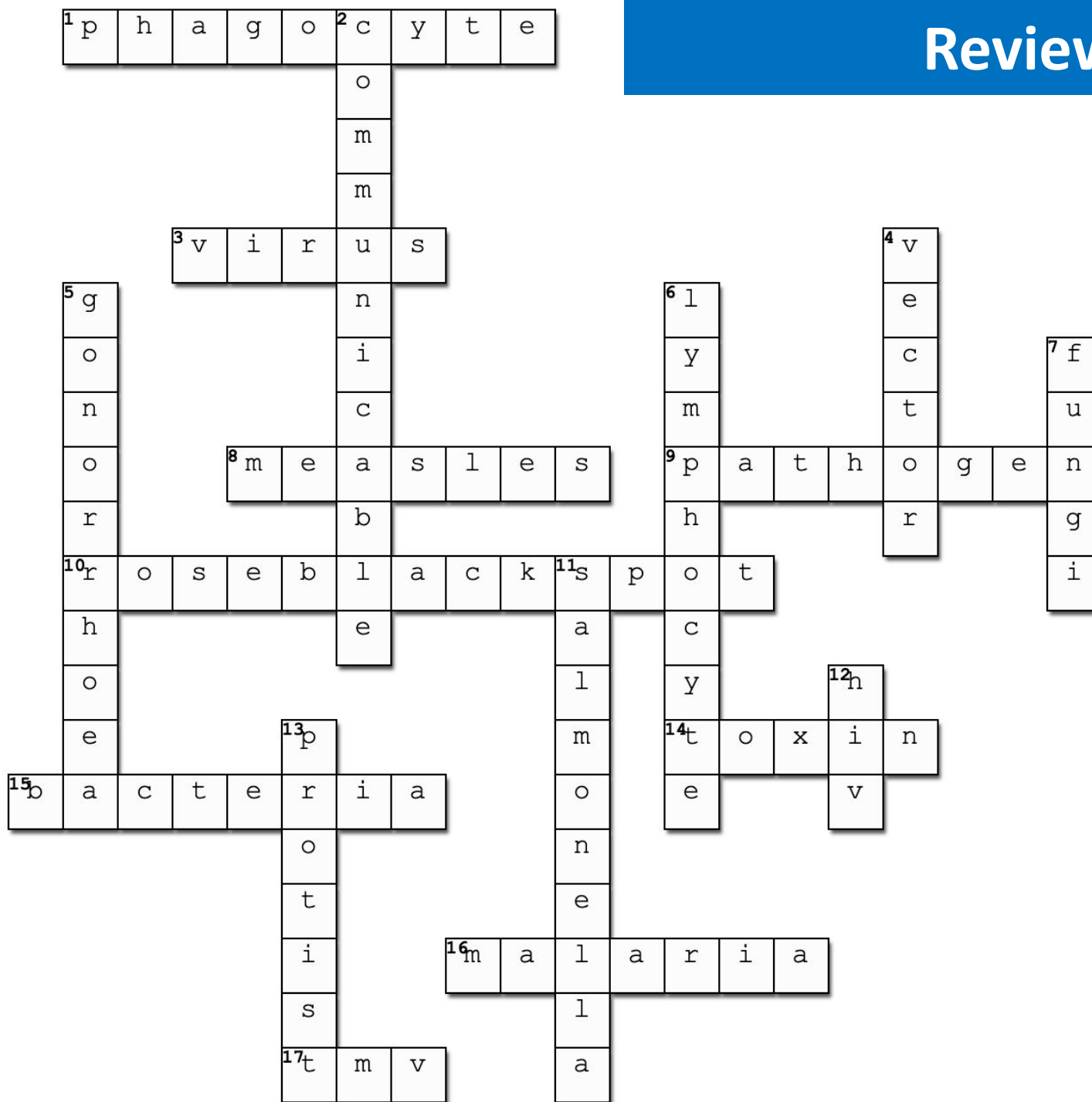
Key Words: Antimicrobial, lymphocyte, phagocytes, antigen, antibody, immune system

Starter: Attempt the cross word about infectious diseases.

EXT: What makes us more likely to get a disease?



Review



Human Defence Responses *

Key Words: Antimicrobial, lymphocyte, phagocytes, antigen, antibody, immune system

Starter: Answer these three review questions below.

1. Name the disease in each plant and the pathogen [2 marks]

Rose plant



Tomato plant



2. Describe how malaria is spread [3]

3. Explain why these diseases in the pictures above are bad for plants [4]

Review

1. Name the disease in each plant and the pathogen [2 marks]



Rose black spot: Fungi



TMV: Virus

2. Describe how malaria is spread [3]

Vector [1] called mosquitos [1] pass protists into the blood stream when mosquitos females feed [1]

3. Explain why these diseases in the pictures above are bad for plants [4]

The discolouration reduces the chlorophyll [1]
Reduction in chlorophyll reduces the rate of Photosynthesis [1], therefore a reduced production of glucose [1] (so) the plant is weaker and growth is reduced [1]

Human Defence Responses *

Key Words: Antimicrobial, lymphocyte, phagocytes, antigen, antibody, immune system

Starter: Let's do a quiz on what we learnt last lesson. 1 □ 10

TIME FOR A

QUIZ!



Quiz:

1. Name the fungal disease that causes this ☐
2. Name the vector for malaria?
3. What is the shared symptom of malaria and measles?
4. Name one way to control the spread of malaria.
5. Describe the environment that fungi like to grow in?
6. What spray is used to kill fungi?
7. How do viruses make us feel ill?
8. What drug works on bacteria but not viruses?
9. What did Semmelweis do and what was the result of him doing it?
10. What disease targets the immune system cells?



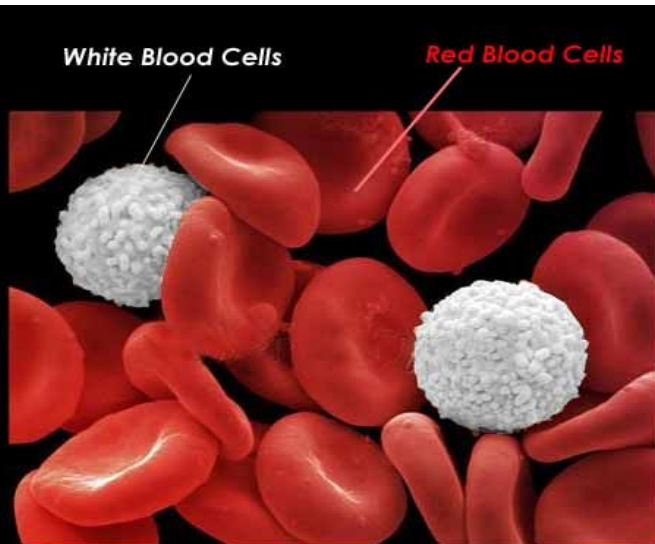
LOb: Understand the different ways the body protects us against pathogens.

Keywords:

Antimicrobial, lymphocyte, phagocytes, antigen, antibody, immune system

Learning Outcomes:

- Identify the different mechanisms that prevent pathogens entering the body
- Describe the role of white blood cells in the immune system
- Explain the different methods WBC can employ to kill pathogens



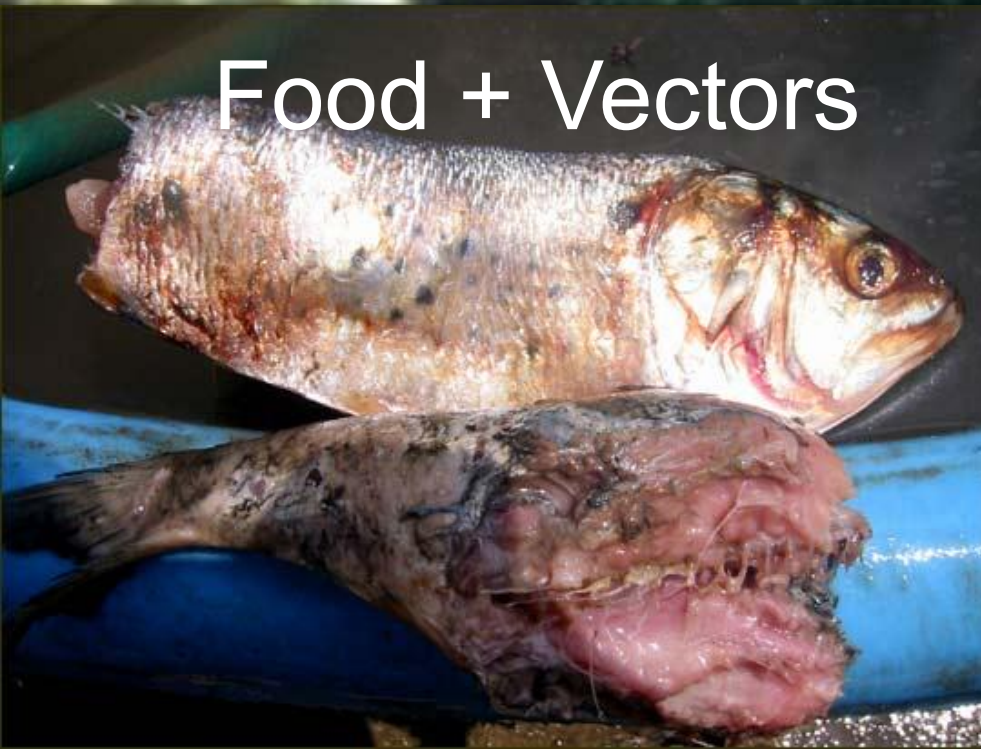
Did you know... in a healthy human there are about 4500 white blood cells (WBCs) per microlitre. Whereas, there are about 5-6million RBCs in the same



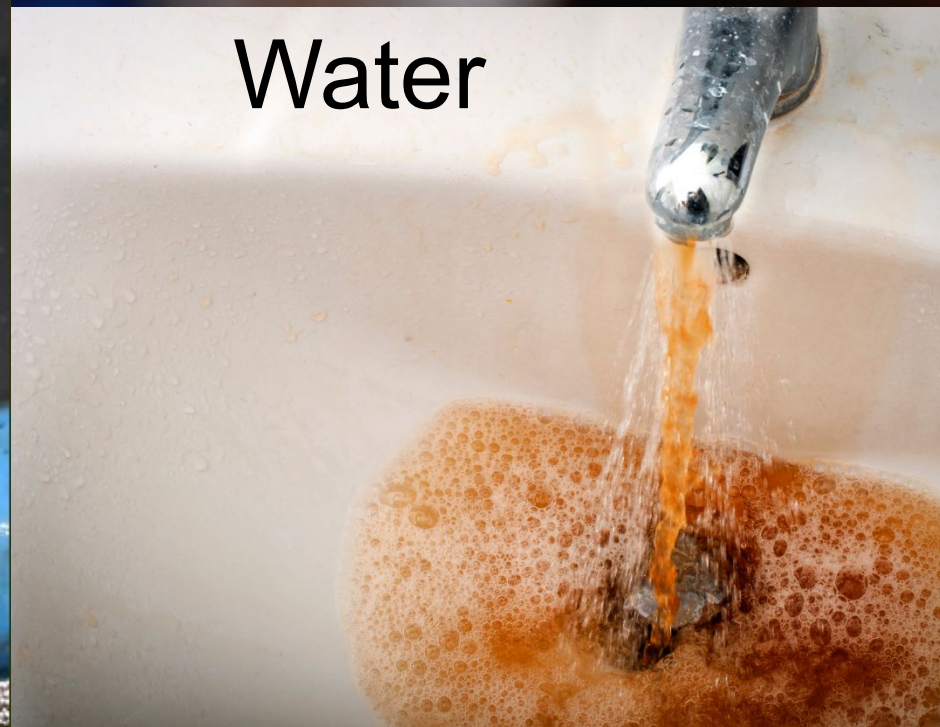
Droplets



Direct contact



Food + Vectors

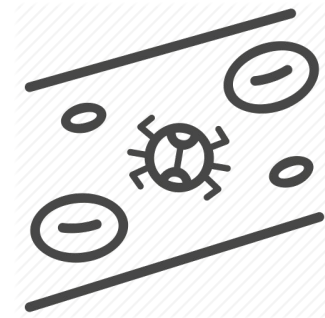


Water



THE FIRST LINE OF DEFENSE

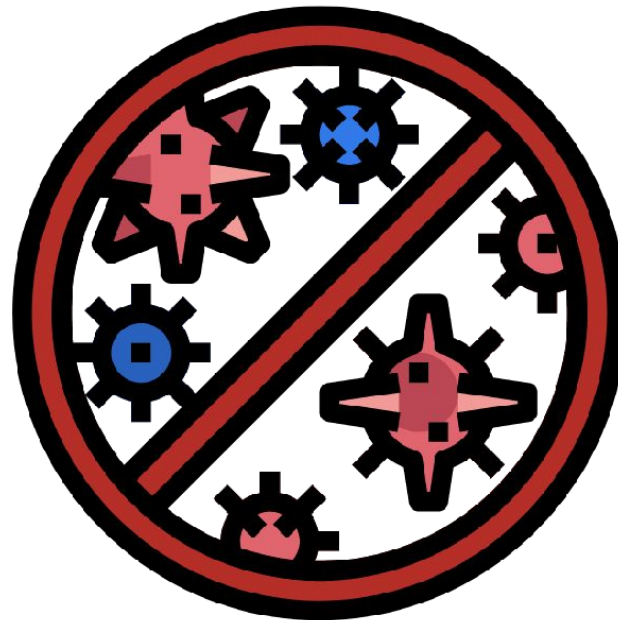
The **skin, digestive system** and **respiratory system** provide a first line of defence to stop the pathogens entering the blood stream.



The body tries hard to stop pathogens entering.
Activity: With the person next to you complete the tasks.

Task 1: List all the places pathogens could enter.

Task 2: State what your body has at those locations to stop pathogens entering.

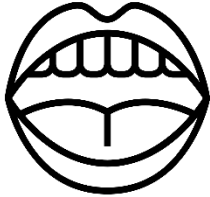


Review

Task 1: List all the places pathogens could enter.



Nose



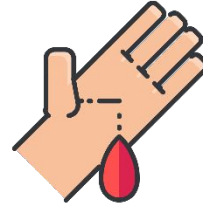
Mouth



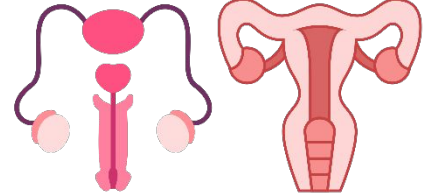
Eyes



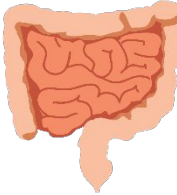
Ears



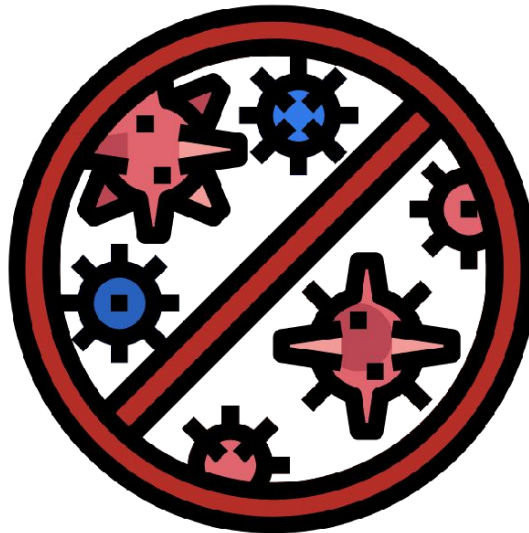
**Cut in
skin**



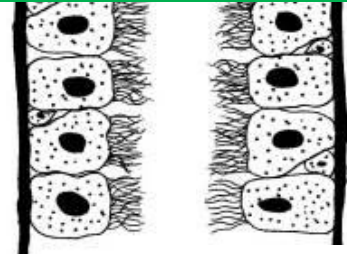
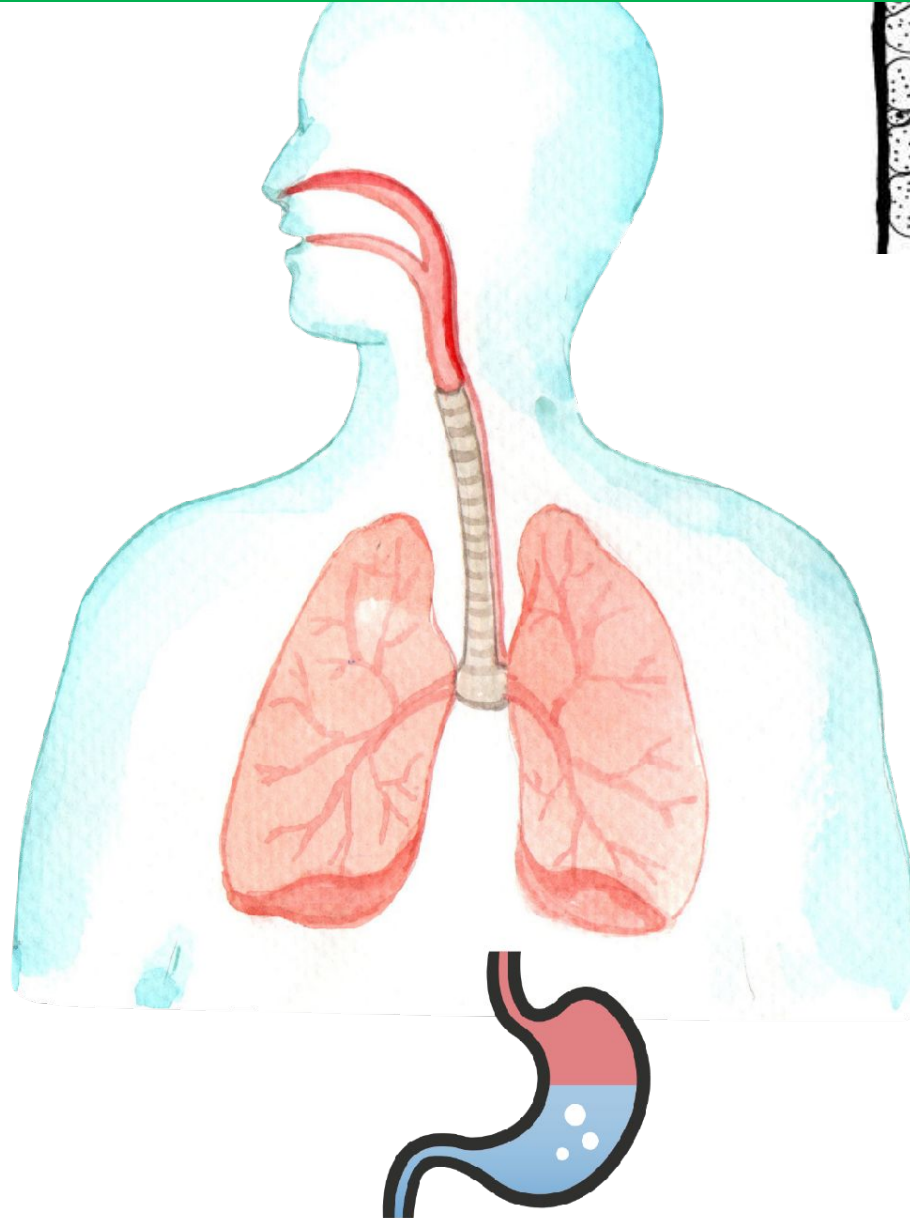
**Reproductive
organs**

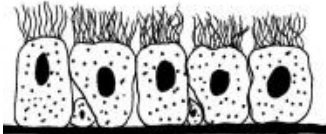


Rectum



First line of defence: Complete the table as we run through these locations.



Location	First line of defence	How it stops pathogens
Nose 		
Trachea / Bronchi 		
Stomach 		
Skin 		

The **respiratory system** is a **weak link** in your bodies defences. Every breath is full of pathogens and when you eat and drink, pathogen get **access to the digestive system**.

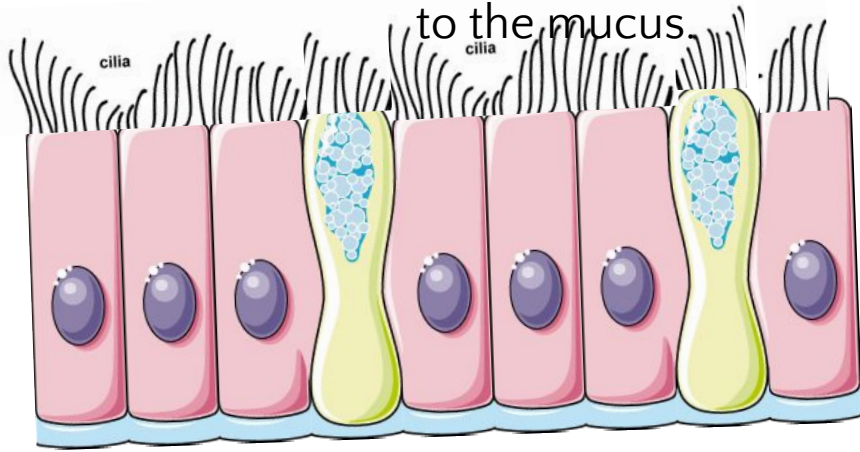
The Nose

Thin hairs and a sticky liquid called mucus trap pathogens



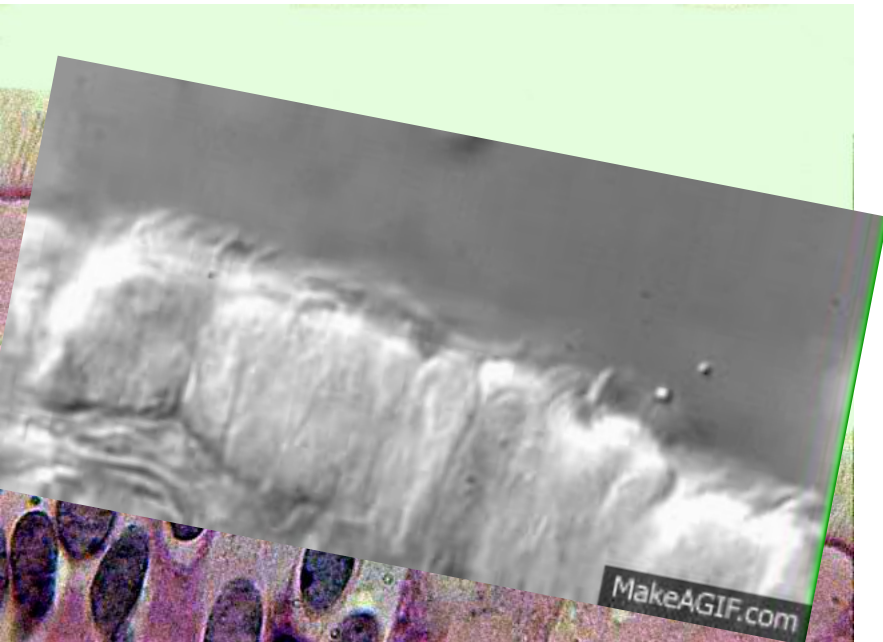
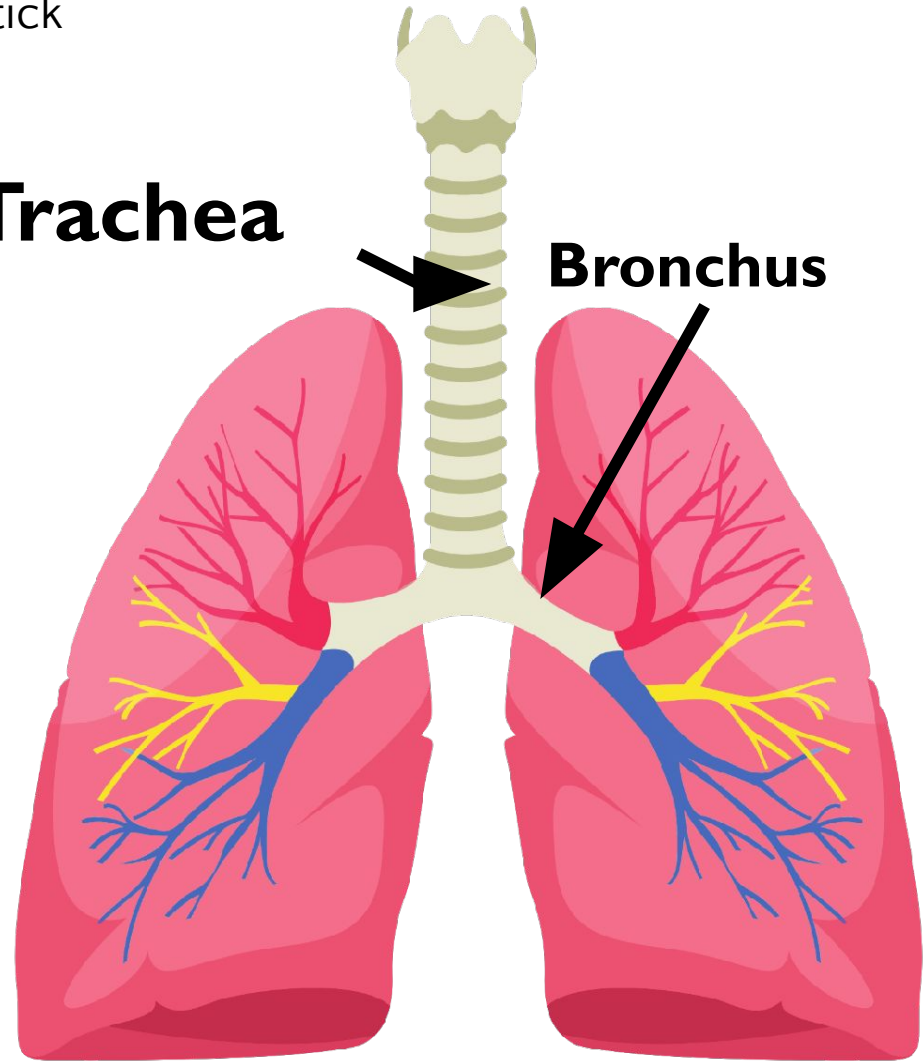
If the pathogens make it past the nose then in your trachea / bronchus there are **cilia**.

Ciliated cells have tiny hairs. They move mucus produced by **goblet cells**. Pathogens stick to the mucus.



Trachea

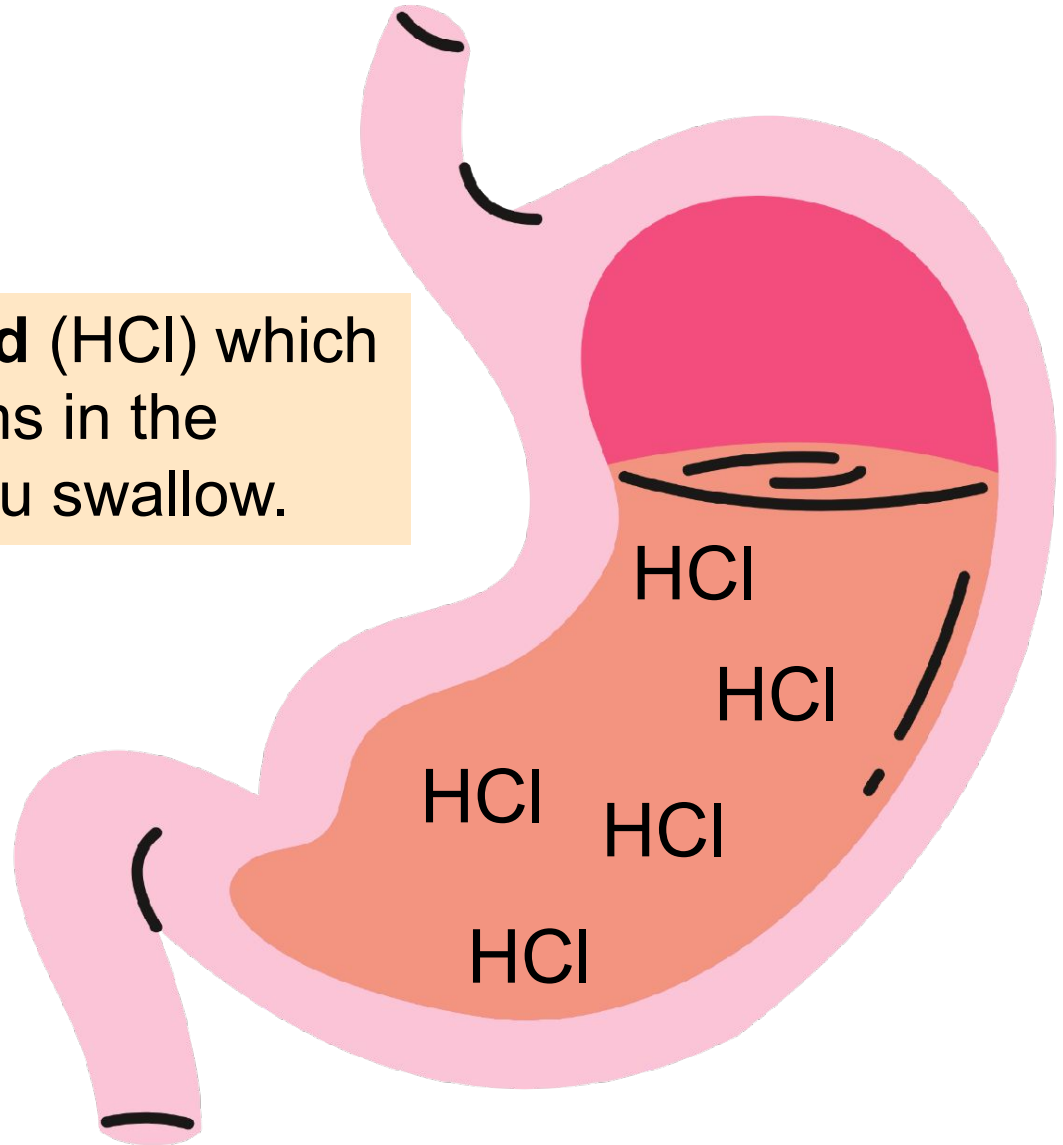
Bronchus



The **mucus** is moved to your throat by cilia to swallow into your stomach.

Stomach

Contains **hydrochloric acid** (HCl) which destroys the microorganisms in the mucus or food and drink you swallow.



The **largest organ** of your body is the skin. It provides a **physical barrier** to pathogens entering the tissue beneath.

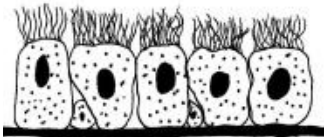
Skin

If the barrier is broken you bleed and **platelets** are involved with forming a **blood clot** (scab) to stop bacteria entering.



Antimicrobial fluid is released that kills bacteria.

Different communities of **microorganisms** also live on your skin which provide an additional **physical barrier**.

Location	First line of defence	How it stops pathogens
Nose 	Thin hairs and mucus	Thin hairs are a physical barrier. Mucus is sticky so traps pathogens
Trachea / Bronchi 	Ciliated cells / hair and mucus produced by goblet cells	Ciliated cells move the layer of mucus to the throat to be swallowed
Stomach 	Hydrochloric acid	Destroys pathogens in food and drink and in air swallowed
Skin 	Skin, antimicrobial fluid, blood clots, communities of microorganism	Physical barrier to pathogens, the antimicrobial fluid kills bacteria

Review

Nose



Hairs and produces mucus

Hairs: filter and physical barrier

Mucus: Sticky traps pathogens

Stomach Acid

Hydrochloric acid produced
that is strong enough to kill
most pathogens



Skin

Blood clots, antimicrobials, microorganisms

Helps to seal up the damaged area

Chemicals that destroy pathogens

Keep skin healthy and provide a barrier

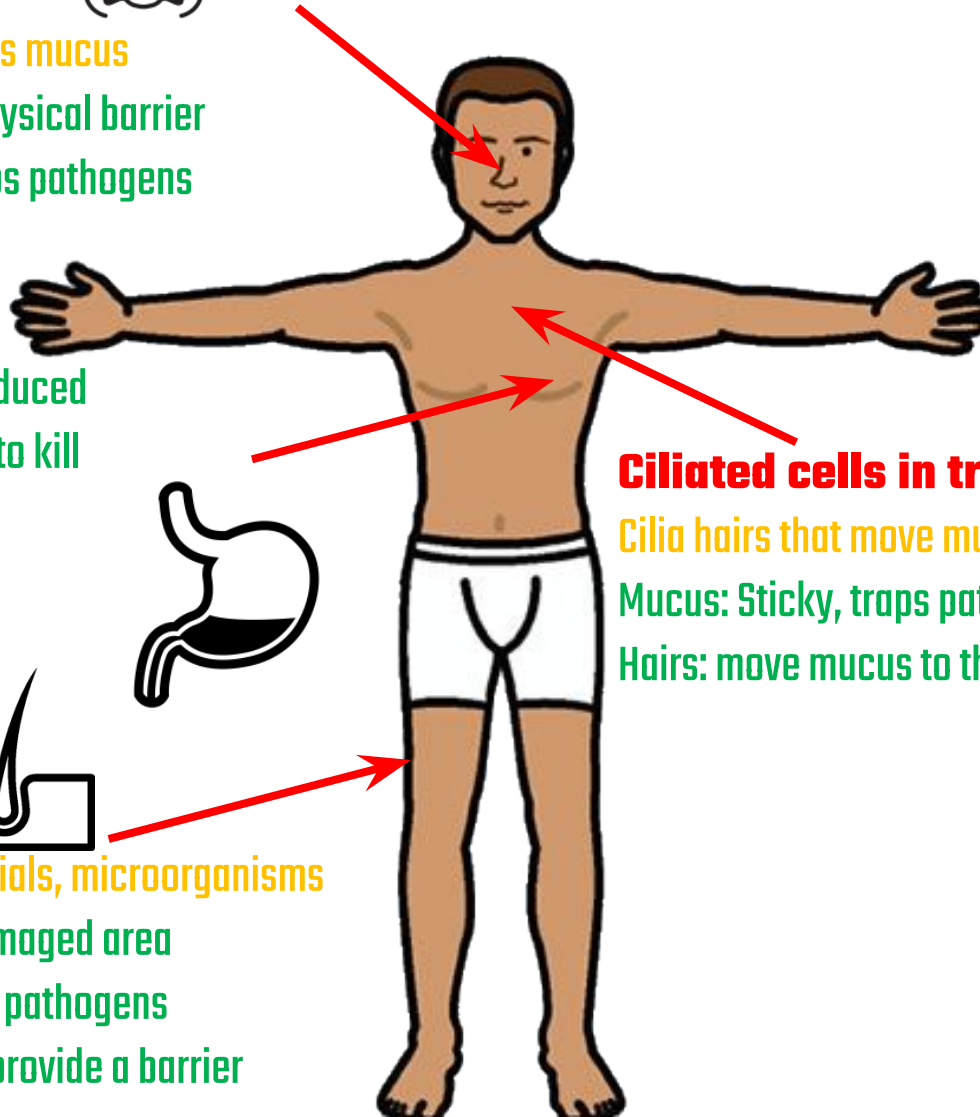
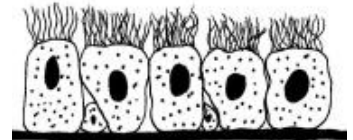


Ciliated cells in trachea

Cilia hairs that move mucus

Mucus: Sticky, traps pathogen

Hairs: move mucus to throat to swallow



physical

cilia

hydrochloric

blood clot

kills

platelets

nose

mucus

mouth

antimicrobial

The first place a pathogen is likely to enter is the _____ and _____. Cells that have thin hairs called _____ have a thin layer of sticky liquid called _____ which traps pathogens and the hairs move it to be swallowed. The stomach contains an acid called _____ acid. This _____ pathogens. The skin is a _____ barrier to pathogens. If there is a cut special cells called _____ go to the wound to form a _____ that closes the wound. A liquid that kills bacteria called _____ fluid is released by the skin. Healthy skin is covered by microorganisms.

The first place a pathogen is likely to enter is the **nose** and **mouth**. Cells that have thin hairs called **cilia** have a thin layer of sticky liquid called **mucus** which traps pathogens and the hairs move it to be swallowed. The stomach contains an acid called **hydrochloric** acid. This **kills** pathogens. The skin is a **physical** barrier to pathogens. If there is a cut special cells called **platelets** go to the wound to form a **blood clot** that closes the wound. A liquid that kills bacteria called **antimicrobial** fluid is released by the skin. Healthy skin is also covered by microorganisms.

Exam Question

Describe the ways in which the human body tries to prevent the influenza virus infecting the body.

(3)

Indicative content	
• nose / trachea have mucus to trap the virus	1
• nose / trachea have cilia / hair to move mucus out	1
• stomach acid will kill the virus	1

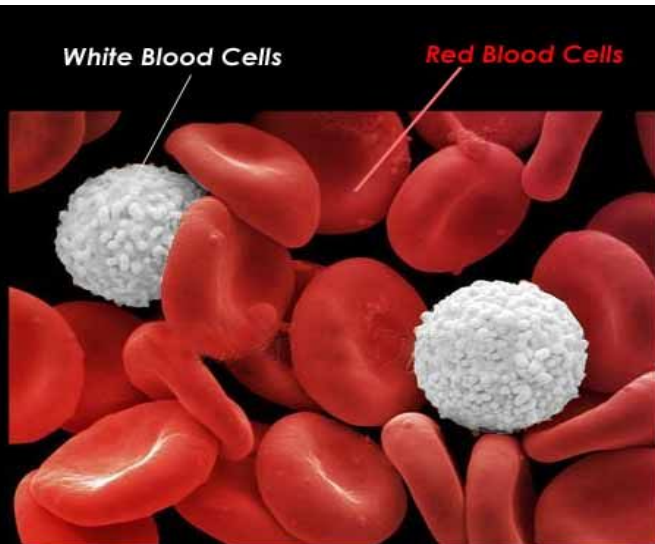
LOb: Understand the different ways the body protects you against pathogens.

Keywords:

Antimicrobial, lymphocyte, phagocytes, antigen, antibody, immune system

Learning Outcomes:

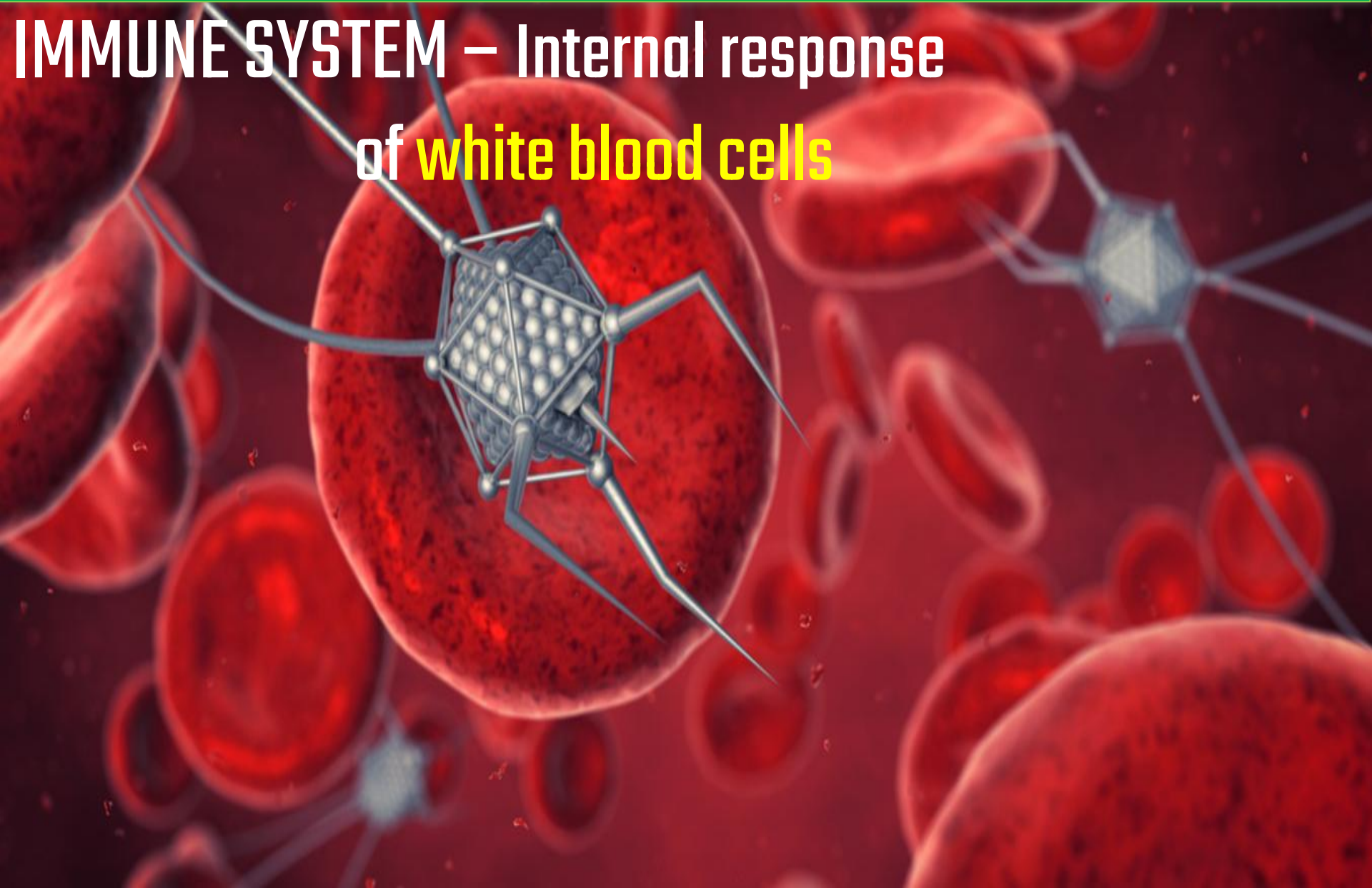
- Identify the different mechanisms that prevent pathogens entering the body
- Describe the role of white blood cells in the immune system
- Explain the different methods WBC can employ to kill pathogens



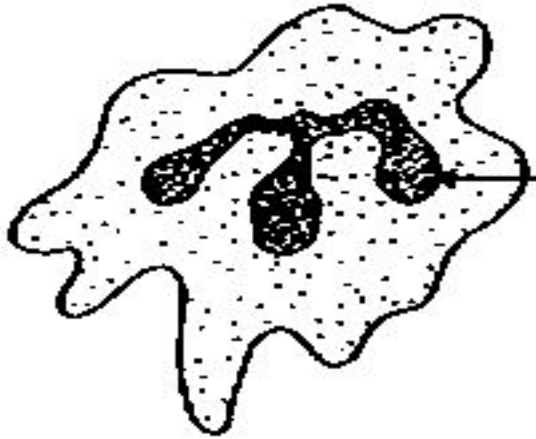
Did you know... in a healthy human there are about 4500 white blood cells (WBCs) per microlitre. Whereas, there are about 5-6million RBCs in the same

What happens if a pathogen makes it past the first line of defence?

IMMUNE SYSTEM – Internal response
of **white blood cells**

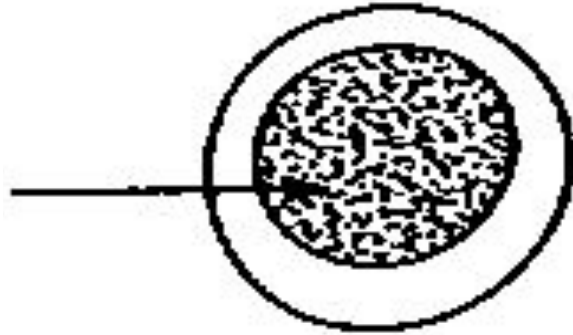


There are many different **white blood cells** but only two you need to be familiar with:



Phagocyte

Nucleus



Lymphocyte

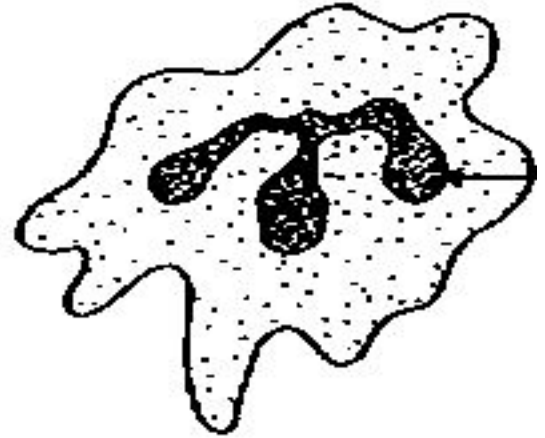
Activity: Watch the video and describe in your own words what you think **phagocytosis** means.

WATCH AND

LEARN

AFTERALL KNOWLEDGE IS

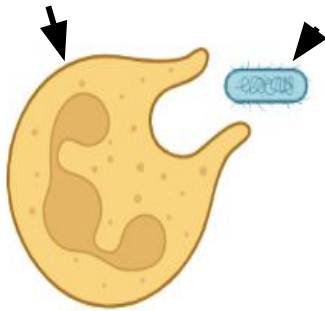
POWER!



Phagocyte

Note. In the video it says macrophages but phagocytes do the same thing

Activity: Fill in your diagram as the video plays.



Response of
the immune
system

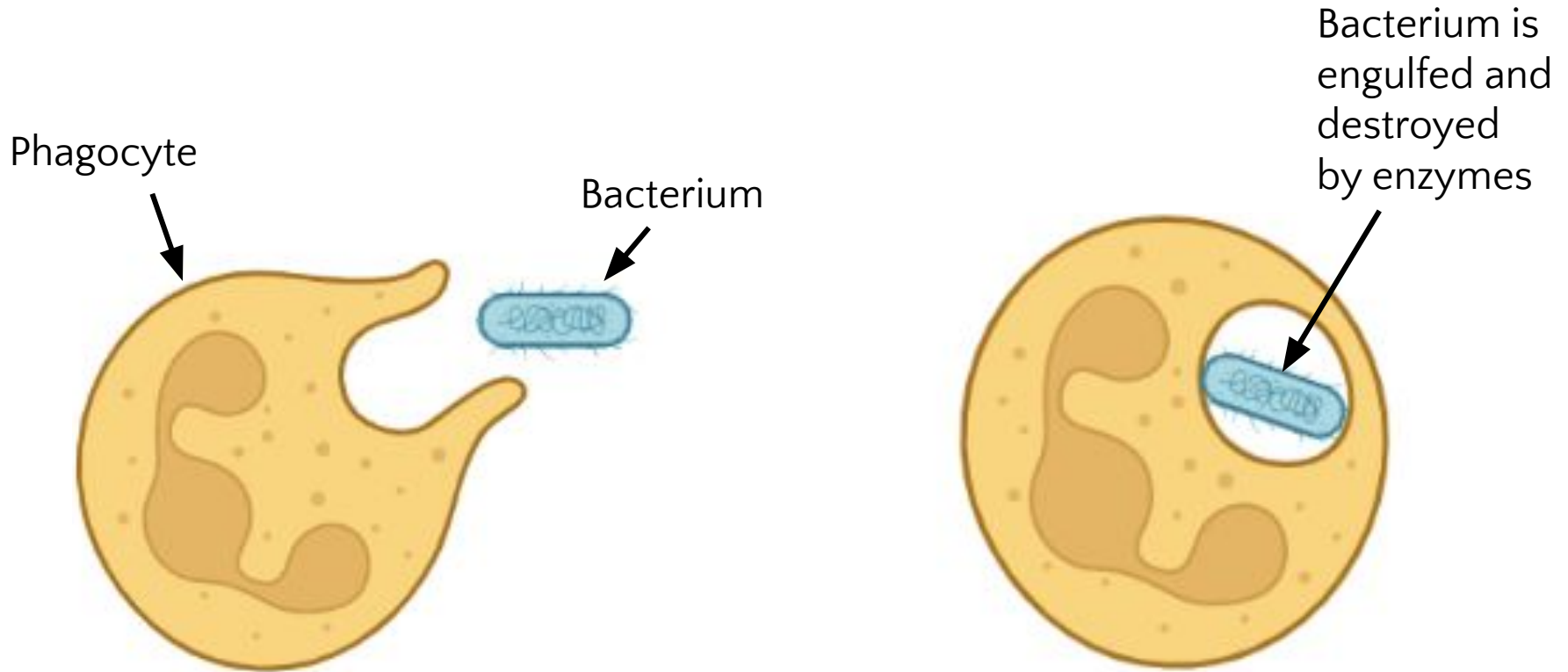
WATCH AND

LEARN

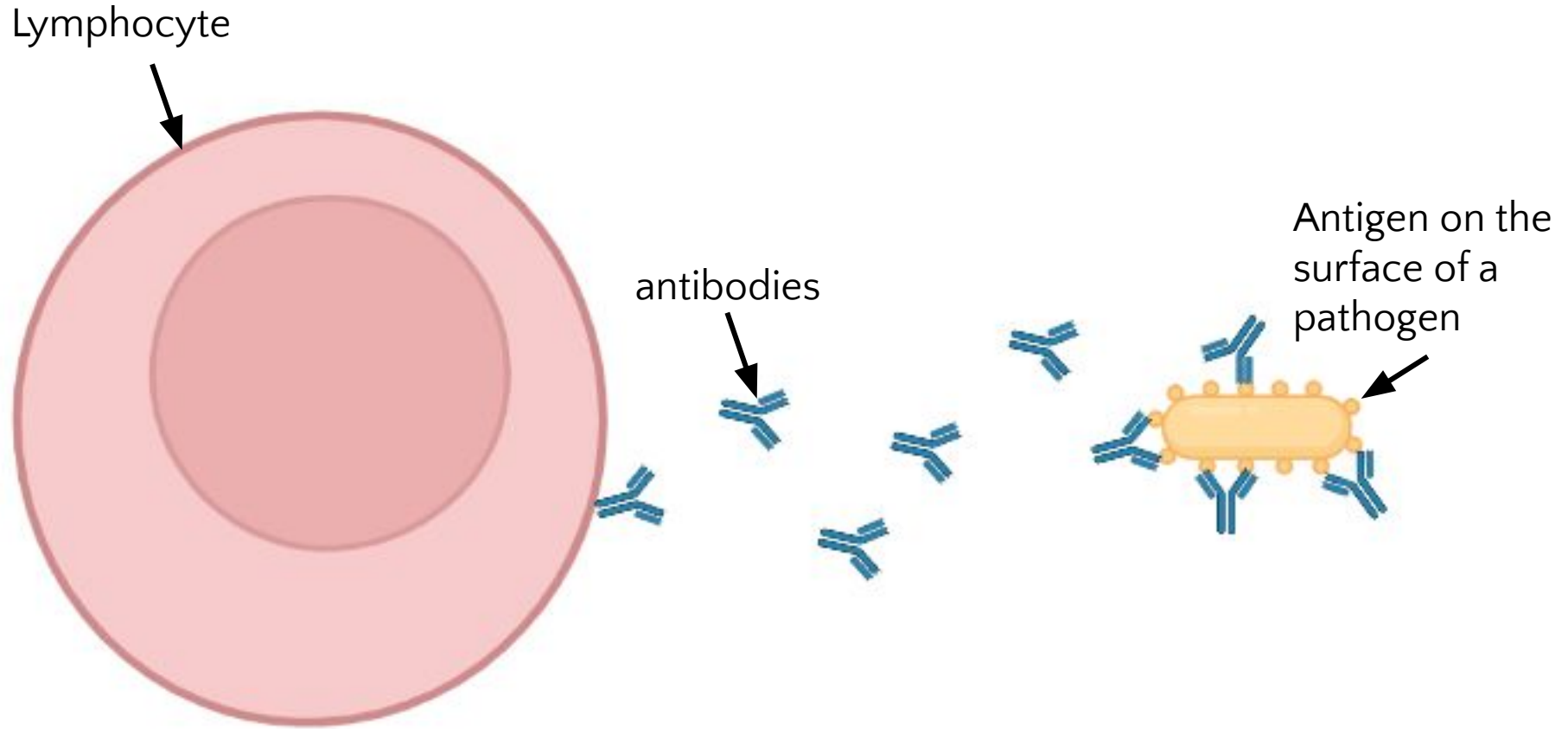
AFTERALL KNOWLEDGE IS

POWER!

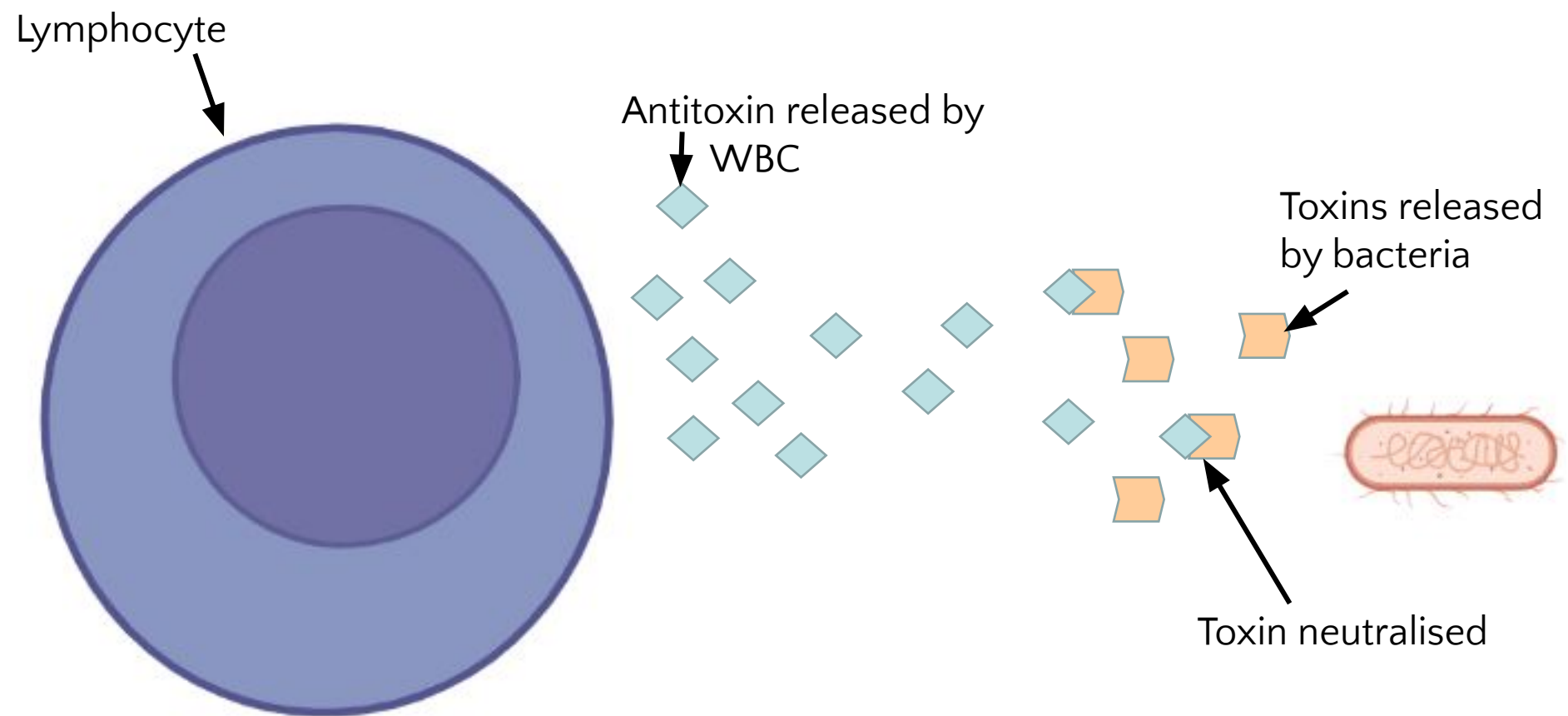
Phagocytosis: means the **ingestion** of **bacteria** or **material** by **phagocytes** (white blood cells)

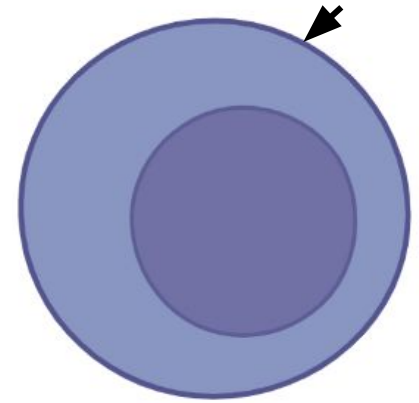
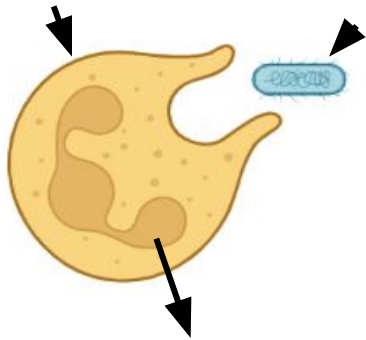


Lymphocytes release specific **antibodies** that **bind** to the **antigens** on the **surface** of the pathogen to **neutralise** it.

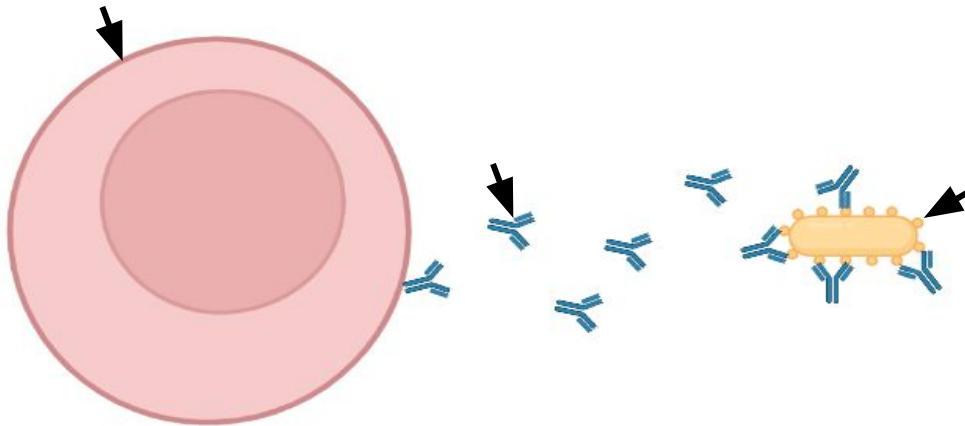
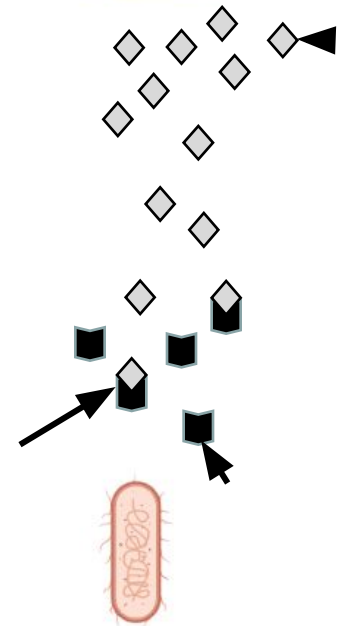


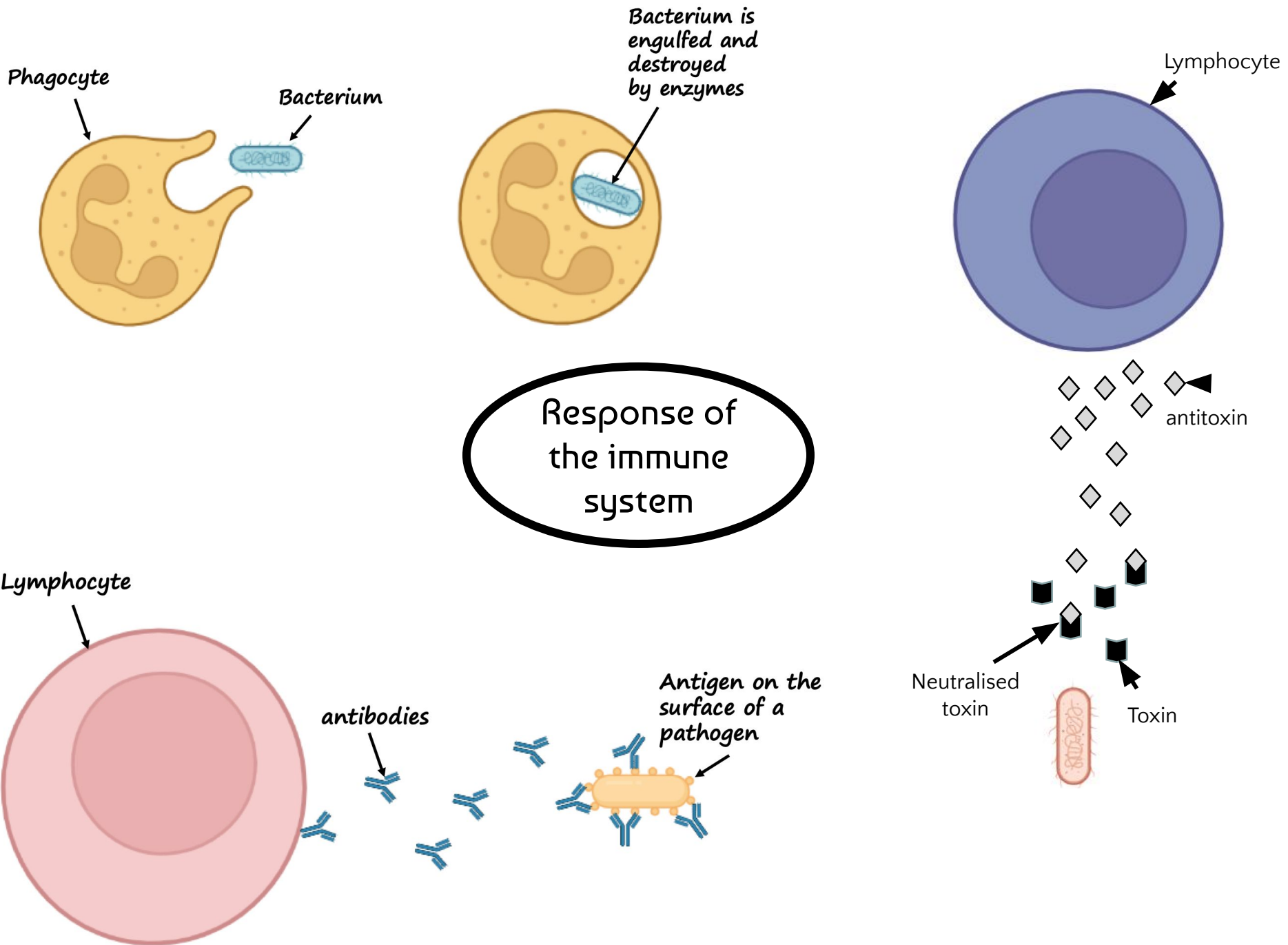
Antitoxins: Lymphocytes release specific antitoxins that bind to the toxins released by bacteria to neutralise their effects.





Response of
the immune
system

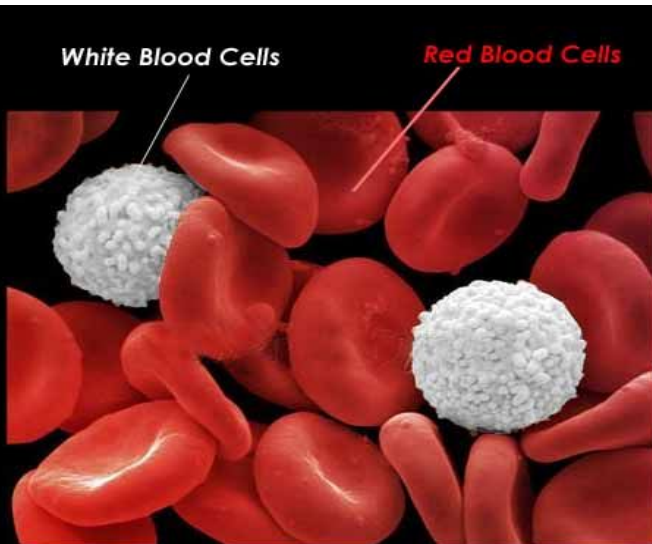




LOb: Understand the different ways the body protects you against pathogens.

Keywords:

Antimicrobial, lymphocyte, phagocytes, antigen, antibody, immune system



Learning Outcomes:

- ✓ Identify the different mechanisms that prevent pathogens entering the body
- ✓ Describe the role of white blood cells in the immune system
- ✓ Explain the different methods WBC can employ to kill pathogens

Did you know... in a healthy human there are about 4500 white blood cells (WBCs) per microlitre. Whereas, there are about 5-6million RBCs in the same

keyfacts

-  The first line of defence includes, **skin, hairs, mucus, wax, stomach acid** and **antimicrobial fluid**.
-  **White blood cells** are part of our internal defence; killing pathogens.
-  Two type of white blood cell include **phagocytes** and **lymphocytes**.
-  Phagocytes do **phagocytosis** and lymphocytes release **antibodies**. WBCs also release **antitoxins**.

Plenary: Make a question

Write two questions that you could ask someone else in the classroom related to what we have studied at the beginning of the lesson and the end.



Be prepared to ask your question

Plenary: Try these practice exam questions.

Practice makes perfect.

